Statistical Modeling of NREL Household Energy with Machine Learning Methods

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Introduction

Background: The rising demand for energy, accompanied with constraints in supply and changing energy policies, shows the importance of understanding residential energy consumption. Since households account for a large proportion of national energy use, identifying how energy is consumed at the household level is critical for finding opportunities to reduce demand and promote efficiency. Therefore, constructing a model for the household energy is critical for estimating the minimum household energy needed.

The U.S. National Renewable Energy Laboratory (NREL)[1] has built a residential energy consumption dataset that captures household characteristics, appliance usage, building design, and other influencing factors. The dataset provides a unique opportunity to analyze energy demand patterns and explore consumption by constructing a model.

Objective: This study aims to develop and evaluate statistical models that can reliably estimate household energy consumption using NREL's residential datasets. By identifying key predictors, the goal is to build interpretable models that improve understanding of energy demand and support smarter, more efficient residential energy planning.

[1] NREL. ResStock database. https://resstock.nrel.gov/datasets

Methodology

The statistical modeling was conducted using machine learning methods as shown in Figure 1.1. Raw data from the NREL database was preprocessed through operations such as data type transformation and outlier filtration. The processed dataset was then randomly split into 70% for training and 30% for testing. The training subset was used to fit the model, while the testing subset was applied to evaluate performance. Model accuracy was assessed using Mean Squared Error (MSE). Hyperparameter tuning (HT) was implemented to optimize the modeling parameters.

Raw NREL database (AZ)
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Pre-processing data
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70/30 Training/Testing split
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Comparative training
↓
- Linear Regression
- Random Forest Regressor
- Gradient Boosting Regressor
- Hist Gradient Boosting
- Light Gradient Boosting
- Neural Network (MLP)

↓
Hyperparameter Tuning
↓
Modeling and Testing

Figure 1.1 Statistical Modeling Pipeline

Results

Cooling Energy Modeling

Table 2.1: Best Model Option with MSE for Cooling Energy Modeling

Best Model Option	Mean Squared Error (kWh²)
Gradient Boosting Regressor	5,638,390.12

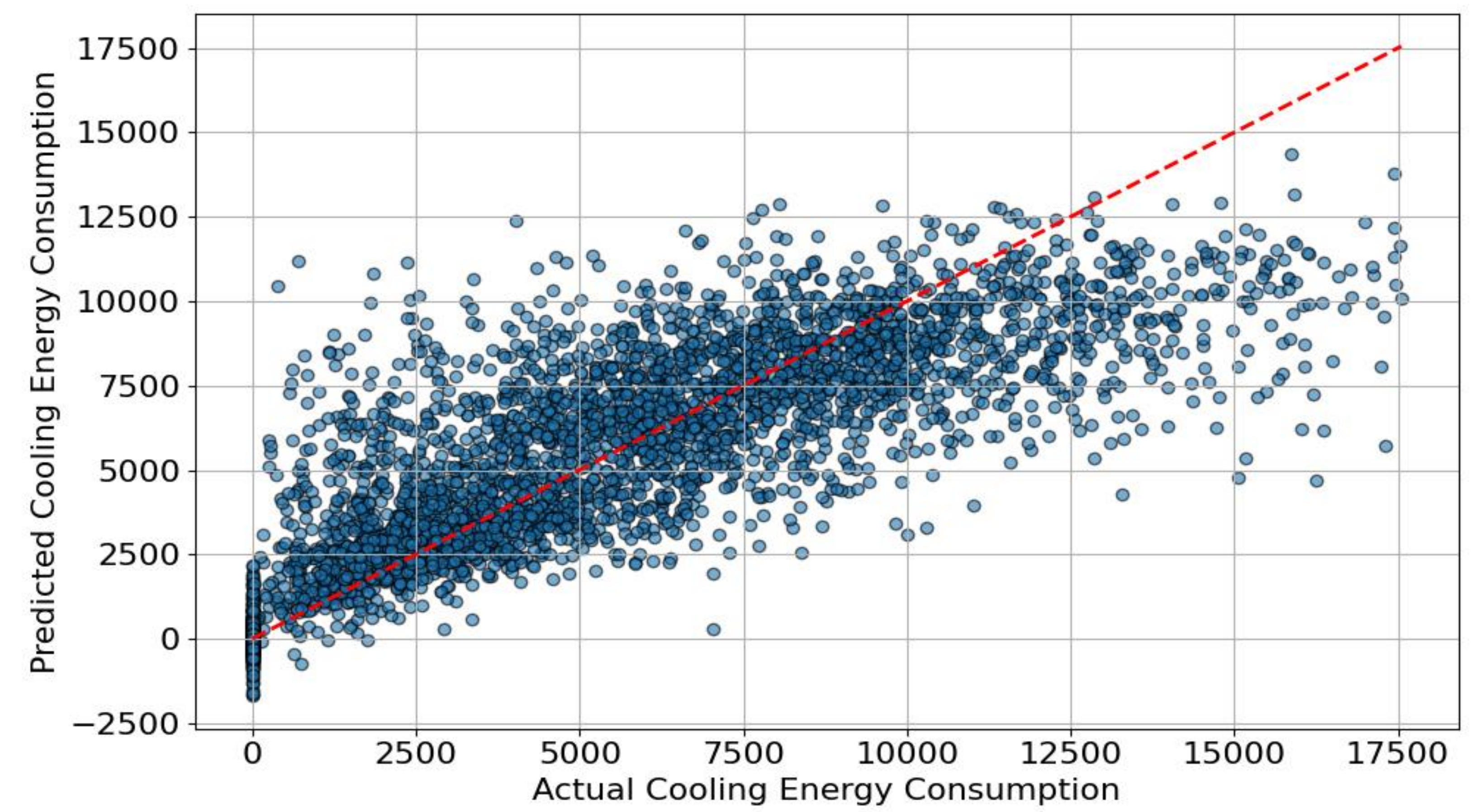

Figure 2.2: Actual / Predicted Cooling Energy with GBR

Table 2.3: Top 5 Influencing Variables with Importance Values in Cooling Modeling (Total IV = 1)

	Variable Name	Importance Value (IV)
1	Building Geometry	0.314
2	Geometry Building Type	0.180
3	AC Cooling Type	0.161
4	Floor Area	0.100
5	Income	0.044

Total Energy Modeling

Table 3.1: Best Model Option with MSE for Total Energy Modeling

Best Model Option	Mean Squared Error (kWh²)
Light Gradient Boosting	20,988,192.43

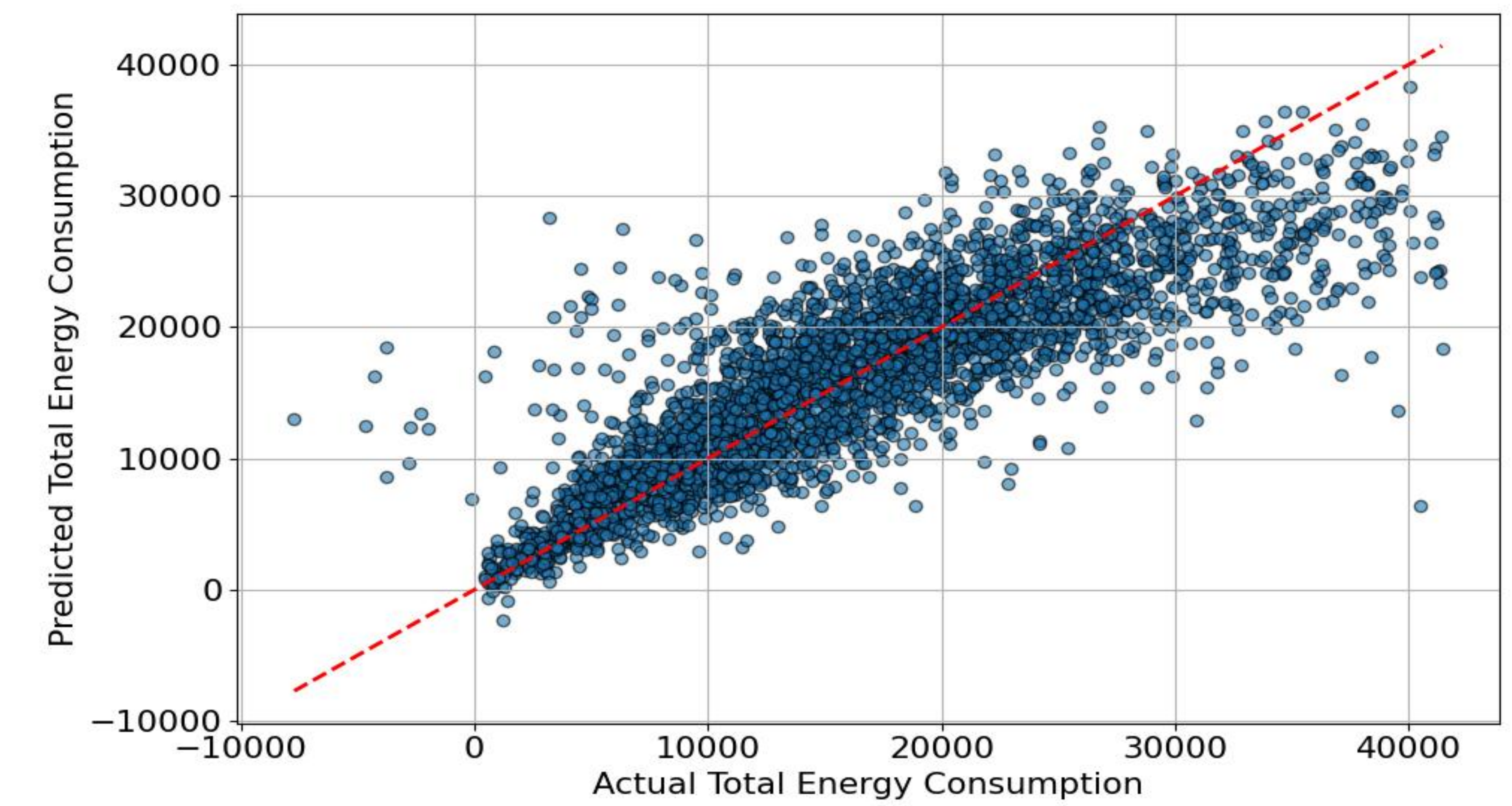

Figure 3.2: Actual / Predicted Total Energy with LGBR

Table 3.3: Top 5 Influencing Variables with Importance Value in Total Modeling (Total IV = 1)

	Variable Name	Importance Value (IV)
1	Building Geometry	0.168
2	Geometry Floor Area	0.093
3	Occupant Count	0.087
4	Vintage	0.080
5	Temperature of Starting Heating	0.074

Discussions and Future Work

For cooling energy modeling, the Gradient Boosting Regressor demonstrated strong predictive performance with a relatively low MSE ≈ 5.6M kWh² as shown in the Table 2.1. The variable importance analysis shown in Table 2.3 suggests that building geometry plays the dominant role, followed by building type, HVAC cooling type, and floor area. These findings follow expectations, as structural design and cooling systems can highly influence the cooling loads. For total energy modeling, the light Gradient Boosting Regressor showed effectiveness with MSE ≈ 21.0M kWh² as shown in Table 3.1. The importance value analysis shown in Table 3.3 suggests that the features in building structure have the greatest contribution to the total energy consumption to the household, which corresponds to the findings in the cooling energy modeling.

For future work, expanding the dataset to include more granular temporal patterns (e.g., hourly or seasonal loads) could improve model accuracy, particularly for total energy consumption. Finally, exploring hybrid models or deep learning approaches may help understand interactions more effectively, enhancing predictive power and enabling real-time energy management applications in smart building systems.

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